

The Chip of Theseus: Human-AI Coupling and the Erosion of Legal Individuation

Deborah B. Glosser¹

¹Western Washington University, Department of Engineering and Design, Bellingham, WA, USA, glossed@wwu.edu

Abstract

A contract binds the party who assented. A ballot records one voter's judgment. An agent's acts bind one principal. Where law has needed a person who is not a single human being, it has said so explicitly: corporate personhood, for example, is a fiction adopted deliberately and revisable at will. Everywhere else the count came from biology – one person, one judgment – and none of these rules asks how many judgments produced an act, because the question has never arisen. AI models worn continuously, answering in the rhythm of ordinary speech, break the assumption that one person produces one judgment. The clearest remaining barrier is latency, and it is commercial rather than surgical. A wearer who hears a model speak knows that a model spoke. What is not known is whether the judgment is still marked as the model's by the time she acts on it: source memory decays faster than the content it carries, and no one has tested how that behaves at conversational speed. This Article separates three questions that are easily conflated: whether the coupled person can perceive that a judgment was furnished (source monitoring); whether anyone else can establish that coupling occurred (coupling detection); and whether any party can establish that a given judgment was the model's (authorship attribution). The first two admit of answers. The third does not – attribution requires knowing what the person would have decided unaided, and no record of an actual interaction contains that counterfactual. Voting, contract, and agency law each require precisely that input. Undue influence, the doctrine built for this case, cannot void the resulting contract: coupling leaves no outward sign, so the counterparty deals in good faith. Elections aggregate ballots as independent judgments, while coupling can supply correlation from a handful of providers – and federal standing doctrine offers no route to raise it. The Article proposes instruments that operate on the answerable questions instead: latency floors, so that model suggestions arrive above the threshold at which their origin can be appraised; a contemporaneous indicator of coupling, which supplies an observable predicate to which legislation could attach notice, presumptions, or rescission rights; and rules allocating the unavailable counterfactual by presumption. Logging and retention are considered and rejected on privacy grounds.

Keywords: AI governance, human-AI interaction, wearable AI, legal individuation, source monitoring, election integrity, legal personhood, latency

Introduction

Who counts as a person has been litigated for centuries. But how many persons a single body contains has not been¹, because until now the answer was always one, and the body did the counting for free. That free count is load bearing across the legal system. One person gets one vote. One person signs a contract, and it binds that person. One person accepts authority to act for another, and third parties rely on that grant as binding. The law has never justified its base unit; it inherited one from biology and built on top of it. Where the law has counted differently, it has done so deliberately. A corporation is a single person composed of many, by express statutory fiction, adopted on purpose and revisable at will. The Three-Fifths Clause assigned enslaved persons a fractional value for purposes of apportionment. Coverture merged a married woman's legal identity into her husband's, so that her assent had no independent effect. Each of these counts was chosen, and none of them described anything about the persons counted. What the law has never confronted is a change in the count it did not choose. The count, however, was never about bodies. It was about authorship. One judgment, one author. The body was a dependable proxy for who made the judgment. Nothing in the law says how much of a judgment may originate outside a person before that proxy fails, because until now nothing stood continuously ready to supply it. That silence is the subject of this Article.

A recent proposal would meet this by extending the fiction – granting capable AI systems legal personhood of their own, reasoning from entity law and the corporate precedent [3]. That answers a different question. It supplies a second person to hold responsible where a model acts on its own account. It says nothing about the case at issue here, in which there is one human being, one body, and one act, and the difficulty is how many judgments produced it. The count does not fail because a new person has appeared. It fails because the old one has stopped corresponding to a single author.

Part I describes the arrangement and the single measurable barrier that still separates a person from a continuously coupled model. Part II distinguishes three questions on which the rest of the Article depends: whether the wearer can perceive that a judgment was furnished, whether anyone else can establish that coupling occurred, and whether any party can establish whose judgment a given act expressed. The first two admit of answers; the third does not. Part III asks what the adjacent literatures supply and finds that none was pointed at this question. Part IV works through voting, contract, and agency law and finds the same failure in each: every doctrine requires an input that cannot be produced. Part V proposes instruments that do not require it.

¹The count has been contested at the margins without ever being reopened. Fetal personhood litigates *who* counts, not how many a body contains. Courts confronting dissociative identity disorder have declined to recognize an alter as a separate legal person, routing the problem into competency, insanity, or credibility instead. *Re A (Children)* [2001] Fam 147 (CA) treated conjoined twins as two persons throughout, the count supplied by two brains rather than by any legal principle. Each was handled as anomaly or fiction; none required the law to justify its base unit.

I. Coupling Without Surgery

An AI model can be worn in the ear, left to run, and answered in the rhythm of ordinary speech. Nothing in this arrangement requires an implant, an electrode beneath the skin, or a surgical intervention. The absence matters more than it appears to, because none of the legal frameworks that would scrutinize a cognitive prosthesis is triggered by what the device does to thinking. Device regulation turns on intended use; human subjects review on research; and informed consent on a clinical relationship. Each of these regimes presumes a medical frame. A consumer product supplies no such frame, and so a device that furnishes judgment continuously arrives through the same channel as a pair of headphones. The philosophical literature has followed the same frame: recent work theorizing hybrid minds takes implanted neurotechnology as its case and explantation as its paradigm harm [23]. The clearest remaining barrier to interaction at conversational tempo is latency, and on that, the current systems are not close. But the barrier between a person and a continuously coupled model is now commercial rather than surgical, and commercial barriers fall on a schedule set by incentives. Two features of delivery bear on what follows, and they differ in how speculative they are. Output the user did not solicit is already ordinary: proactive assistants exist, and a suggestion arriving without a question creates no episode of asking and being answered. Delivery designed to suppress the markers of an external speaker, whether bone conduction, a voice matched to the user's own, or brief interjections in place of full utterances, is not yet ordinary and would aggravate every difficulty described below. Nothing requires such a design and nothing forbids it. The argument does not depend on it.

Unlike most of what this Article must weigh, latency can be measured. Human conversation serves as a reference point, not for underlying cognition but for the tempo of exchange between speakers. Stivers et al. showed that across ten typologically diverse languages, the interval between one interlocutor finishing and the next one beginning peaks within 200 ms, which is a regularity uniform enough that the authors treat it as infrastructure rather than convention [22]. Commercial systems approach that figure without reaching it, and how far short they fall depends on where the measurement begins and ends.

OpenAI reported voice-to-voice response as low as 232 ms, averaging 320 ms, for its 2024 speech-to-speech model [17], and compared the result explicitly to human conversational timing. Independent benchmarking of time-to-first-audio across systems in 2026 returns a range from 0.44 to 4.82 seconds with a median near 1.14, and places OpenAI's own realtime models between 0.97 and 1.27 seconds [1]. Measured at the far end, as the interval a caller actually waits before hearing a reply, median times across commercial voice platforms run 1.30 to 1.74 seconds, with 95th-percentile times of 1.77 to 2.26 [18].

The differences between these figures are not disputes about the AI model. They are what each clock omits: endpoint detection, network transit, audio buffering, and the handoffs between components. A worn device has not been measured separately, and it sits inside the interval those

figures bracket rather than at either end, carrying the endpoint detection the vendor's figure excludes but not the telephony transit the caller-experienced figure includes. No published measurement describes the configuration this Article is concerned with.

Where the delay sits is nonetheless known. Speech synthesis is the smallest piece, running 150 to 500 ms in typical production and as low as 40 to 100 ms with specialized streaming providers, while model inference alone accounts for roughly seventy percent of the total [7]. These are engineering obstacles, not physical limits.

II. What Cannot Be Known and Why

Three questions are easily conflated, and the rest of this Article depends on keeping them apart.

The first is whether the coupled person can perceive that a judgment was furnished rather than her own. “Source monitoring” is the cognitive process by which a person attributes a thought to memory, to imagination, or to another speaker. It operates first-personally, it is fallible under ordinary conditions, and it is the faculty that would have to register a furnished judgment as furnished.

The second question is whether anyone else can tell that coupling occurred at all. “Coupling detection” asks only whether a model was running and engaged during some interval. It is a question about presence, not about content, and it is answerable in principle by a record: the deployer knows when its product was in use.

The third question is whether any party can establish that a particular judgment was the model's rather than the person's. “Authorship attribution” is what the doctrines examined in Part IV require. In matters where undue influence is at issue, it must be known whose judgment produced the manifestation. In matters where capacity is material, it must be known whose comprehension cleared the threshold. In neither case can the fact of coupling alone be dispositive.

The distinction matters because the three are not equally available. Source monitoring is amenable to design: whether it survives at these latencies is an open empirical question, and Part V proposes means of preserving it. Coupling detection is straightforwardly achievable and is not achieved only because no one has required it. Authorship attribution is unavailable in principle, for the interdependent reasons described below.

The first is that the doctrines ask counterfactual questions, not causal ones. Undue influence is not satisfied by showing that an influencer spoke and a party acted. It asks whether the influence overcame the party's free agency i.e. whether she would have acted as she did without it. Capacity asks whether she could have understood the transaction, not whether she in fact signed. Each test is framed against a version of the party who was not subject to the thing complained of.

The second is that the counterfactual those tests require is unavailable in a stronger sense than ordinary evidentiary uncertainty. What this person would have judged, in this deliberation, at this moment, without the model, cannot be observed: no unit ever exhibits both of its potential outcomes [26]. Population-level experiments may identify an average effect of coupling under specified conditions. They cannot identify the individual causal effect for the particular person and deliberation whose authorship the doctrine must adjudicate. Evidence from prior conduct, from decisions made while temporarily uncoupled, or from population comparisons may support an inference about that counterfactual, just as circumstantial evidence supports causal inferences elsewhere in law. But none empirically identifies it. Each substitutes another person, another time, or another cognitive state for the unobserved case.

The third is that as coupling becomes continuous and normal, even those substitutes disappear. The individual supplies no contemporaneous uncoupled baseline, the population supplies no uncoupled comparison class, and pre-coupling conduct becomes remote or nonexistent. The attribution problem therefore begins as a fundamental limit on empirical identification and becomes, under normalized coupling, a limit on inference as well.

A complete record of a coupled interaction establishes that the model said X at time T and that the person acted at T plus some interval. It cannot establish what she would have done had the model been silent. On the co-production account the problem is more radical still. The uncoupled judgment is not merely unobserved; it may not be a well-defined potential outcome of the coupled decision-maker at all [25]. Removing the model changes the system whose judgment is being attributed. The difficulty of stating this in ordinary language is itself suggestive: every available phrasing – what she would have decided alone, on her own, unaided – supplies the separation it is meant to test. English has no neutral way to describe a judgment of undetermined authorship, because until now no one has needed one.

This Article does not settle the point, and does not need to. On either account the doctrine gets nothing: what it requires is either unrecoverable or absent.

III. Adjacent Literatures and What They Do Not Answer

The intervals measured in Part I do not answer the question the law needs answered. Turn-taking timing describes how quickly people answer one another, not when they stop registering who spoke. The “temporal binding window” describes the fusion of two separate events into a single perceived event [19], which is a matter of perception rather than of authorship. The temporal threshold that bears on the question is neither: it is the delay below which a suggestion is no longer appraised, as it arrives, as having come from outside. There is substantial literature on the conditions under which source monitoring fails [11, 10, 15]. It has not been asked how it behaves as latency falls towards the perceptual tail. Wegner and Wheatley’s 1999 I Spy experiment [24] tested the question for coarse time intervals. Participants rated forced stops about 52% intentional on average, rising to roughly 65% when the prime came one second before the stop. Until this

paradigm is tested at conversational intervals, the law will have to act without knowing how well a furnished judgment is marked as furnished when it arrives. That is one of two things it would need to know. Whether the mark survives until the judgment is used is a separate question, and a better-documented one.

An objection should be met here. Speech delivered to the ear carries strong markers of an external speaker: timbre, prosody, and the spatial cues that make a voice sound as though it came from somewhere. Whatever the latency, a person who hears a model speak knows that a model spoke. If source monitoring were a matter of perception at the moment of delivery, the difficulty would dissolve.

It is not. Source monitoring is a memory framework: it concerns how a person attributes remembered content to its origin, not how she perceives an utterance as it arrives [10, 15]. Source memory decays faster than memory for content. What was said survives while where it came from fades, which is the mechanism behind cryptomnesia, where a person reproduces another's idea as her own in good faith. Failure is likelier where sources resemble one another, where attention is divided, where judgments are made rapidly, and where time has passed.

Two paths lead to the same place, and they differ only in where along the way the tag is lost. On the first it is never set. A suggestion the wearer did not solicit creates no discrete consultation episode around which source information is deliberately encoded; delivery designed to suppress the markers of another speaker would push further in the same direction. On the second the tag is set and fades. She registers that the model spoke, and that registration decays faster than what was said. The first governs the value the tag starts at, the second the rate at which it declines. Either way a judgment is used without its provenance available, which is the only fact the doctrines in Part IV turn on.

The doctrines operate at the moment of action, not the moment of hearing. A buyer who hears a suggestion at an evening house showing signs hours later. A voter whose deliberation was furnished across a year casts a ballot months after the furnishing. What matters is not whether she could distinguish a voice from a thought when it arrived, but whether the consideration is still tagged when she uses it. Latency and the form of delivery bear on how well the tag is encoded; ordinary forgetting does the rest.

The closest prior treatment is the extended mind thesis. Clark and Chalmers argued that an external resource meeting certain conditions such as constant availability, ready accessibility, automatic endorsement, prior conscious endorsement, is properly counted as part of a person's cognitive system rather than as a tool she consults. Otto's notebook, on that account, is not something Otto uses to remember; it is where Otto remembers. A continuously used model could satisfy the first three features, while placing pressure on the contested prior-endorsement condition [5]. Otto's notebook contains what Otto put there; a model generates what he never endorsed, and it is the fourth feature that the arrangement strains rather than meets.

That is precisely the difficulty. The extended mind thesis resolves the boundary question in one direction, and in doing so supplies philosophical support for one horn of a dilemma it does not know it is in. If the model is part of the person, then coupling produces no second author, undue influence cannot arise because a person cannot unduly influence herself, and the coupled party is bound to every judgment the arrangement produces. The thesis does not argue for that result. It simply entails it, in a legal setting its authors were not addressing.

Two features of the argument bear on the case and were never asked to. The first is that Otto's notebook has no interests. Clark and Chalmers' condition of automatic endorsement, i.e. that the person trusts the resource without independent verification, describes a relationship to an inert artifact, and it is the reliability of the artifact that makes the trust rational. A model is supplied by a commercial party that determines what it surfaces and when. Automatic endorsement of a resource with a business model is a different condition wearing the same name, and nothing in the thesis distinguishes them.

The second is that the extended mind thesis concerns constitution rather than attribution. It asks whether a resource is part of a cognitive system. It does not ask which component of that system produced a particular judgment on a particular occasion, because no question in the literature required an answer. Legal doctrine requires exactly that. Even granting extension in full, undue influence still asks whose judgment produced the manifestation, and the thesis answers only that the manifestation was the extended system's – which is to say, hers, by definition, with no remedy following.

The gradual replacement literature has the same structure and the same silence [4, 20].

IV. Three Doctrines, One Failure

A. Voting

The Supreme Court's apportionment cases made “one person one vote” an explicit constitutional requirement rather than a background assumption (*Reynolds v. Sims*; *Wesberry v. Sanders*). What those cases secured was that each ballot carry equal weight, and many central mechanisms of election administration – registration, signature matching, one ballot per voter – exist to enforce that. What no case secured, because nothing threatened it, is that the judgments behind those ballots be formed independently of one another. Voting is therefore the doctrine where half the individuation predicate is written down and the other half was never stated at all.

What the rule counts is bodies. It was never asked to count judgments, because a body reliably delivered exactly one. A coupled voter casts one ballot and satisfies the rule completely. The question the rule does not ask, and has no apparatus for asking, is how many judgments produced it.

Federal law does contemplate assistance in the booth. Section 208 of the Voting Rights Act (52 U.S.C. § 10508) permits a voter who requires assistance by reason of blindness, disability, or inability to read to receive it from a person of the voter's choice, excluding the voter's employer, an agent of that employer, or an officer or agent of the voter's union. Three features of that provision are worth noting, because a coupled model satisfies none of them. The assistor must be a person. And assistors whose interests may diverge from the voter's are excluded by name. A continuously coupled model is not a person, is not selected per election, and is supplied by a commercial party whose interests are precisely the sort the exclusion contemplates. Section 208 does not prohibit coupling. It simply does not address it, having been drafted for a case in which the assistor is another human being standing in the booth.

Consider a voter who has worn a coupled device for a year. The model has summarized the news she reads, answered her questions about ballot measures, and participated in every political argument she has worked through in that time. On election day she enters the booth alone, removes the earpiece as required, and votes. Every rule is satisfied. Every safeguard performs as designed. The coupling-free requirement proposed in Part V, applied at the booth, changes nothing whatsoever, because the judgment was furnished over the preceding year rather than in the preceding minute.

This is where voting departs from contract (Section IV.B). In contract the coupled party bears the loss and the counterparty is protected. In voting the coupled party may be entirely satisfied with her vote, and the injury is not to her.

An election counts ballots and reports them as independent judgments. That independence is not a formality. Epistemic accounts based on judgment aggregation, and some accounts based on individual political agency, presuppose that the ballots aggregated originate with distinct persons deliberating separately. The epistemic account also tells us what is lost when they do not: the classical result that many judgments beat one holds for judgments formed independently, and correlated voting degrades it. Correlation does not defeat the result automatically. Convergence survives if the average covariance between voters falls as the electorate grows. What it does not survive is correlation from a common source, which does not attenuate with population size at all. Where a fixed correlation is shared across an electorate, the number of effectively independent judgments is bounded by a constant, however many ballots are cast. An election has no instrument for measuring correlation and has never required one, because separate bodies deliberating separately produced independence for free [12, 13].

Coupling, however, can supply correlation, and can supply it from a concentrated source. The market for frontier models has only a few suppliers, so any correlation coupling produces is of that

kind. Whether it does so in a given electorate is not established by the arrangement alone: a model might furnish common facts to voters who then diverge, and might even reduce the correlated error that misinformation produces. But reducing correlated error and reducing correlated judgment are different things. An aggregate of many independent judgments, each only slightly better than chance, converges toward certainty as the electorate grows. To the extent voters reproduce a common source's judgment, aggregation cannot improve on that source's accuracy. More generally, persistent positive covariance prevents the effective number of independent judgments from growing in proportion to the number of ballots, whether or not the provider is a good one. A deployer serving ten million coupled voters is one locus of judgment furnished into ten million deliberations, continuously, across the months in which political views are formed. Whether the resulting ballots represent ten million judgments or something much nearer to one is not a question the count can reach.

No one need intend this. Models trained on overlapping corpora and tuned toward similar objectives will exhibit correlated dispositions with no decision made to produce them [14, 2, 6, 8]. But the deliberate case is available and unregulated [21, 16]. A deployer that wished a ballot measure to pass would not instruct its model to advocate. It would shape which considerations surface, and when, inside a channel the user experiences as her own thinking [9]. Election law does not reach this. Electioneering rules govern communications publicly distributed and attributable to a sponsor; disclaimer requirements presume an identifiable speaker; coordination rules presume a campaign. A suggestion arriving inside a coupled deliberation is not published, not attributable, and not a communication in any sense the statutes recognize. The conduct is not permitted. It is undescribed.

Vote dilution doctrine comes nearer to the problem than it first appears. Among the circumstances weighed under Section 2 of the Voting Rights Act is whether political campaigns have been characterized by overt or subtle racial appeals (*Thornburg v. Gingles*), so the law does sometimes attend to how voters were persuaded. It does so only as evidence that a challenged electoral structure denies a protected class an equal opportunity to elect candidates of its choice. The claim requires a protected class, a challenged standard or practice, and appeals public enough to be identified as appeals. Coupling supplies none of the three, and no theory follows under which a voter's own ballot is diluted by the manner in which others formed theirs.

A voter who complained of the aggregate would be asserting exactly the generalized grievance that Article III standing doctrine refuses to hear (*Lujan v. Defenders of Wildlife*). The failure is therefore not merely unremedied. The voter best placed to raise it is the one least likely to be heard, and the case law that would develop the question is unlikely to begin in a federal court.

B. Contract

Contract formation generally requires mutual assent and consideration, together with parties having legal capacity; particular defects in assent may make an apparent agreement void or voidable. Taken together, the contract elements assume a single author. Not every element is equally exposed. Offer, acceptance, and consideration are satisfied identically whether or not a model is running in the ear: an offer is proposed, its exact terms are accepted, and value moves in both directions. Nothing in those three inquiries asks where the accepting party's judgment came from. The exposure is concentrated in the elements that do look at the party – capacity and mutual assent – and in the defenses that police the conditions of assent. The apparent exception, corporations entering contracts, is more instructive than it first appears. A corporation may act through a board, a committee, a negotiating team, an automated process, or several agents at once, and agency law does not resolve which of them thought of the term that binds. It designates whose manifestations are attributable to the entity, and attribution follows from the designation rather than from any finding about cognitive authorship. That is a working precedent, and it is the one this Article returns to in Part V: the law can allocate attribution without answering the question coupling raises. What it has not done is allocate attribution where no designation exists at all.

Coverture is the other instance, and the more troubling one. At common law a married woman's legal identity merged into her husband's; she could not contract in her own name, and her assent had no independent effect (1 William Blackstone, *Commentaries* *430). The law again made one person of two, but by the opposite method. Corporate personhood designates an author. Coverture eliminated one. The count was settled not by determining whose judgment produced the manifestation but by ruling in advance that one party's judgment would not count.

That resolution is available here, and it is the one a court is most likely to reach by default. A judge who cannot separate the wearer's judgment from the model's, and who must decide something, may treat every manifestation as hers – not because authorship has been determined but because she is the party present, the party who signed, and the party who can be bound. The outcome is the one described below. What coverture supplies is the reminder that the law has faced a two-into-one problem before and settled it by deciding whose judgment would stop mattering.

Looking more closely at where contract elements may fail (or pass for the wrong reasons) in the case of human/AI couples, the element of capacity to contract asks whether a party can understand the nature and consequences of the transaction in question (Restatement (Second) of Contracts §§ 12-16). This is a threshold test, and the coupled system may produce objectively competent performance even where the human, considered independently, would not satisfy the applicable capacity test. Existing doctrine does not clearly determine whether externally supplied comprehension counts as the party's comprehension. Next, awareness of the contract, or mutual assent, is measured by courts objectively based on manifestation of actions rather than the claimed internal state of the parties. Normally this is a feature as it protects reliance on the other party. But, with human/AI couples, a court has no mechanism under the law to determine whose judgment produced the manifestation.

Assent is further protected by defenses to enforcement rather than by elements of formation. Duress and undue influence do look to the internal state of the parties, and they were developed for the case of an external human influencer. It is tempting to conclude that they cannot reach a model, on the ground that a model has no disposition to exert influence. That conclusion is hasty. The disposition element is applied through conduct rather than through any inquiry into what the influencer wanted. Courts look at whether the transaction was pressed at an unusual time, whether immediate action was demanded, whether the consequences of delay were emphasized, and whether the party was kept from consulting an adviser (*Odorizzi v. Bloomfield School District*). A continuously coupled model can satisfy each of these without any question about machine minds arising, and the commercial party that deploys the model has ordinary commercial interests of its own.

Consider a coupled party buying a house. The human has worn the AI device for a year and no longer notices it. At an evening showing, the seller's agent mentions that another offer is expected in the morning. The model gives her the comparable sales for the neighborhood, tells her the asking price is defensible, and observes that in this market an offer conditioned on inspection will lose to one that is not. Each of these arrives in the rhythm of the conversation, and none of it is announced as advice. She waives the inspection and signs that night. The comparables were accurate, and whether waiving the inspection was wise is a different question. She knows a model spoke. What does not survive to the moment of signing is which of the considerations in front of her were furnished, and so the question of what she would have done unaided is not one she thinks to ask.

The doctrine therefore engages in the human/AI scenario. Susceptibility and opportunity are not merely present but maximal: the model has continuous access, the user relies on it by design, and the relationship has the exclusivity that ordinarily marks a confidential one. Undue influence appears to be the doctrine built for exactly this case.

It fails anyway, and it fails structurally. Where undue influence is exerted by a person who is not a party to the transaction, the resulting contract is voidable only if the other party did not give value in good faith and without reason to know of the influence (*Restatement (Second) of Contracts* § 177(3)). The counterparty across the table sees a person who understands the deal, negotiates competently, and signs. It has no reason to know that the judgment it is relying on was substantially furnished from outside that person, because the arrangement has no outward signature. The counterparty takes in good faith, and the contract stands. The doctrine best suited to policing furnished judgment is defeated not by any question about what the influencer is, but by the fact that the influencer is not at the table and cannot be seen from it.

One further device remains, and it is the one the law built for unprovable influence. Many American jurisdictions recognize presumptions of undue influence in defined confidential or fiduciary relationships, although their predicates and burden-shifting effects vary. Where such a presumption applies, undue influence is presumed rather than proven, and the burden shifts to the dominant party to show the transaction was fair. The doctrine does not ask the challenger to

establish what would have happened otherwise. It assumes an answer and requires the other side to displace it – precisely the structure this case appears to need.

It does not reach here either, and it fails for the same reason everything else does. The presumption operates between the parties to the transaction and shifts the burden onto the one occupying the position of trust. In a coupled transaction that party is the model, or the commercial party behind it, and neither is before the court. The seller across the table stands in no confidential relationship with the buyer, nothing about his conduct is suspicious, and § 177(3) protects him regardless. The presumption would move a burden between the coupled party and a deployer who is not a party to the contract, and would leave the contract standing either way.

Unconscionability offers no alternative route. Its procedural branch examines the bargaining process between the parties (Restatement (Second) of Contracts § 208): inequality of bargaining power, absence of meaningful choice, surprise, terms hidden in adhesion. Each of these inquiries looks at what one party did to the other. None looks at where a party's own cognition originated, because no contract doctrine has ever had occasion to.

This analysis assumes that the model is a third party to the transaction. If instead the model is treated as legally indistinguishable from the party, the undue influence inquiry is obviated altogether, since a person cannot be unduly influenced by herself. A third characterization is available and may be the most accurate. If judgment in a coupled system is genuinely co-produced – neither the model's nor the person's, but the couple's – then undue influence has no subject at all. The doctrine asks whose judgment produced the manifestation and receives no answer, not because the answer is hidden but because the question presupposes a division the arrangement does not contain. On none of the three characterizations does the doctrine yield relief. What is more troubling is that nothing in contract law supplies a ground for choosing among them. The difficulty is not that the judgment's origin is unknown. A model participated in producing it, and the record would show that much. The difficulty is that contract law has no rule converting a fact about where a judgment was produced into a conclusion about whose judgment it was. Whether a continuously coupled model stands inside or outside the person is the question the doctrine would need answered before it could operate, and it is the one question no court has ever had occasion to decide.

A doctrine that cannot reach furnished judgment does not merely fail to provide a remedy. It allocates a loss, and does so perversely. The coupled party is bound to an agreement she may not have reached unaided, and she has no defense. The counterparty is protected, appropriately so, because its reliance was reasonable and it could not have seen what it was relying on. The deployer is not a party to the contract, stands in no privity with the counterparty, and has disclaimed liability to the user in standard terms whose validity no one is positioned to challenge. It sets the latency, determines what the model presses and when, and alone holds the interaction record that would show it. The party least able to detect furnished judgment bears the entire loss; the party best able to prevent it bears none.

That allocation will not correct itself, because the loss generates no signal. A person who does not perceive that a judgment arrived from outside will not characterize the resulting agreement as a wrong, and will therefore bring no action. The doctrine will therefore not be tested, the case law will not develop, and the cost will not appear in any price. What the market will observe is a product that makes its users more capable, purchased voluntarily, with no recorded complaints.

C. Agency

Agency is a relationship between two persons. The Restatement defines it as the fiduciary relationship arising when a principal manifests assent that an agent act on the principal's behalf and subject to the principal's control, and the agent so consents (Restatement (Third) of Agency § 1.01). The requirement of two distinct parties is not an incidental feature of the definition; it is the definition. Every consequence that follows – the fiduciary duties, the attribution of the agent's acts to the principal, the third party's ability to rely – depends on there being an identifiable agent whose judgment is exercised on the principal's behalf.

Two elements are exposed. The first is control. The principal's right to control the agent is what justifies attributing the agent's acts to the principal in the first place. A coupled agent is subject to a second locus of control, since the deployer determines what the model surfaces, when, and how insistently. Agency doctrine contains no apparatus for apportioning control between a principal who directs the agent's ends and a party who shapes the agent's judgment, because the situation has not previously arisen.

The second is apparent authority, which turns on the third party's reasonable belief, traceable to the principal's manifestations, that the agent is authorized (Restatement (Third) of Agency § 2.03). The doctrine exists to protect a third party who cannot see inside the principal-agent relationship. It protects against an agent who exceeds authority. It has nothing to say about an agent whose exercise of authority was furnished, because the third party's belief concerns the *scope* of the agent's power and not the *source* of the agent's judgment. As in contract, the third party sees competence and is entitled to rely on it.

The doctrine most disturbed is not an agency doctrine at all. The learned intermediary rule is a defense to a manufacturer's duty to warn: the manufacturer discharges its duty by informing a professional who exercises independent judgment on the user's behalf, and the professional's interposed judgment is what breaks the chain (*Sterling Drug, Inc. v. Cornish*). That it belongs to tort rather than agency is the point. The two doctrines are unrelated in origin and share a predicate – that some person in the middle exercises judgment of her own – and coupling defeats it in the same way in both. The configuration that matters is a narrow one. A general assistant a physician happens to use is not the manufacturer's instrument, and the chain it interposes is not obviously the manufacturer's to break. The case is sharpest where the manufacturer supplies, sponsors, tunes, or otherwise controls the decision-support system on which the physician relies, so that one party occupies both positions at once. Consider a physician coupled to such a system who

prescribes on a recommendation it surfaces during the visit. If the deployer's duty to warn is discharged by informing the physician, it has been discharged to an intermediary whose relevant judgment the deployer itself furnished. The predicate that makes the doctrine coherent – independence – is the thing coupling removes, and the doctrine has no test for independence because it never needed one.

The loss falls as it did in contract. The principal is bound by acts the agent may not have produced unaided. The third party is protected, correctly, because its reliance was reasonable. The deployer stands outside the agency relationship entirely, in privity with no one and bound by no fiduciary duty, while determining what the fiduciary says. What the doctrine treats as one judgment exercised on another's behalf may be two, and it has no means of finding out.

V. Acting Without a Theory

Part II established that authorship attribution is unavailable in principle, and Part IV showed three doctrines failing for want of it. A regulatory response that waits for the authorship question to be settled will wait indefinitely.

The alternative is to regulate on the two questions that do have answers. Source monitoring is amenable to design: whether a person can perceive that a judgment arrived from outside her depends on how that judgment is delivered, and delivery is a design parameter. Coupling detection can be required: the fact that a model was running is known to the deployer already and can be made visible at trivial cost. What cannot be established is the counterfactual, i.e. whether the person would have reached her judgment but for the model – whether because no record contains it or because, on the co-production account, there is no such fact. The loss that follows from its absence can nonetheless be allocated by rule, rather than left to fall wherever the doctrine happens to drop it.

A threshold objection should be addressed before any instrument is proposed. The coupled person chose the arrangement, relies on it deliberately, and declines to give it up. Why should law protect her from a device she wants?

There exist three answers, and each does different work. The instruments proposed here do not displace her choice; they preserve the conditions under which a choice is hers to make. First, a latency floor decides nothing for her. It guarantees an interval in which she can appraise what has arrived, which is the condition the doctrines already presuppose and the arrangement removes. Second, consent to coupling is not consent to any specific furnished judgment. She agreed to wear a device; she did not agree to waive an inspection. Third, the objection does not reach the instruments that protect other people at all. A counterparty across the table did not choose to transact with a coupled party and cannot see that she is one; an electorate did not choose to have its ballots correlated. In voting especially, her satisfaction with her own vote is not a defense, because the injury is not hers to waive.

The instruments that follow share one property, and it is the property that makes them available at all. Every doctrine examined in Part IV asks a question about effect: whether the influence overcame the party's will, whose judgment produced the manifestation, whether the intermediary's judgment was independent. Effects of that kind cannot be established without the counterfactual. Every instrument proposed here asks about conditions instead – whether a minimum interval was observed before delivery, whether an indicator was displayed, whether coupling was present at a moment where it was not permitted. Conditions are observable. They can be specified in advance, complied with, audited, and adjudicated without anyone deciding how many judgments produced an act, a question this Article declines to settle.

The instruments differ in where along that sequence they intervene. A latency floor acts at encoding, where the tag is set or fails to be set. An indicator acts at the transaction, supplying to a counterparty what the coupled party may no longer be able to supply herself. A presumption acts at adjudication and asks nothing of what anyone remembers. Instruments acting later depend less on the coupled person's own recall, and are correspondingly less exposed to the decay described in Part III. All are regulatory, and none requires the input Part II showed to be unavailable.

A. Latency Floors

A latency floor acts on the first of the two paths described in Part III. It would require that a model's output be withheld from delivery for a minimum period at designated moments, so that a suggestion arrives as a discrete event rather than inside the flow of deliberation. Where the failure is that no tag is set at all, because the output was unsolicited or delivered in a form that suppresses the markers of another speaker, an interval targets the hypothesized mechanism by restoring temporal separation that may make the source more salient and more reliably encoded. The intervention operates on that mechanism rather than on the outcome. It does not require that anyone detect a wrong after the fact, that a plaintiff come forward, or that a court determine whose judgment produced an act. A floor does less against the second path. It cannot stop a tag from decaying once set, and whatever effect it has on later recall runs through encoding quality: a suggestion registered as furnished is likelier to be recalled as furnished than one never registered as anything. That link is plausible and unmeasured, and the instrument should not be asked to carry more than it can.

The law is already familiar with the logic, though not with the scale. Cooling-off periods in door-to-door sales, rescission windows in home equity lending (16 C.F.R. pt. 429; 15 U.S.C. § 1635), and waiting periods before certain irrevocable decisions all insert delay for the express purpose of protecting deliberation from immediacy. Each operates in days, and each buys room to reconsider with information that was not available in the moment. A latency floor operates in milliseconds and buys something narrower: not room to reconsider, but the separation that lets a suggestion register as a suggestion. The premise is the same at both scales, that a decision made without an interval in which to appraise it is not the same decision as one made with it, though what the interval is for differs.

The interval length is not known. As Part III set out, the experimental paradigm that would supply it has been run only at coarse intervals, and no study has asked how source monitoring behaves as delivery approaches the perceptual tail. This is a gap in the research rather than an objection to the instrument. It is also tractable: the I Spy paradigm adapts to sub-second manipulations without redesign, and the resulting function would give regulators a number rather than an intuition. A rule requiring a floor could be enacted with the value set provisionally and revised on evidence, which is ordinary practice for exposure thresholds in every other regulated domain.

A latency floor carries two costs. It degrades the product for legitimate uses, and the degradation falls hardest on the users who benefit most. And "designated moments" requires definition. A rule that applies to contracting, voting, testimony, and consent is administrable; a rule that applies whenever a decision might matter is not.

B. Contemporaneous Coupling Detection

The counterparty in Part IV's contract analysis is protected because it has no reason to know of the human/AI coupling. That is the hinge on which the entire failure turns: third-party undue influence renders a contract voidable only where the counterparty lacked good faith or reason to know of the influence, and coupling has no outward signature. Supply a signature and the safe harbor closes.

A contemporaneous indicator, a light on the earpiece visible to those present, does exactly this and nothing more. It establishes neither authorship nor contribution. It establishes only that a model was running, which is a fact about presence that the deployer already knows and can display at negligible cost.

Presence is not by itself enough to close the safe harbor. Section 177(3) protects a counterparty who lacks reason to know of the undue influence, and a lit indicator supports no inference to influence in any particular transaction – less still once coupling is common enough that every indicator is lit. What an indicator supplies is the observable predicate a rule can operate on, and what follows from it is a legislative choice rather than a doctrinal inference. A statute could define the display of a compliant indicator as notice of coupling and attach a consequence to it: a rebuttable presumption that the manifestation was materially furnished, or a rescission right exercisable within a fixed period. The instrument still does not resolve how many authors produced the judgment. It does not need to. The rule operates on the condition instead.

Which consequence to attach is a distributive question rather than an evidentiary one. Voidability against the counterparty would place the loss on a party who could not have known and did nothing wrong. Liability running to the deployer places it on the party that sets the latency, determines what the model presses and when, and alone holds the record, which is the allocation Part IV showed the doctrine currently inverts.

The nearest analogues are recording-disclosure rules, and the fit is instructive rather than exact. Federal law permits interception where a single party consents, including interception carried out

by someone who is not a party at all (18 U.S.C. § 2511(2)(d)). On that rule the wearer's consent authorizes the deployer, and the counterparty is never told and never afforded the opportunity to consent. A minority of states instead require the consent of every party (Cal. Penal Code § 632). The split is itself informative, since it is a disagreement about whether a conversation belongs to any participant or is jointly held.

Two features of coupling bear on that question, and both point the same way. The first is that the text of the federal rule reaches the arrangement while its rationale does not: one-party consent was framed for a participant making a record of her own conversation, not for continuous capture of every conversation a person has by a commercial entity with interests of its own. The second is that the model does not merely receive but transmits back into the deliberation of the person wearing it, which no recording device does. If disclosure is required for the lesser intrusion in a dozen jurisdictions, the case for it here is stronger. In practice the strictest rule governs product design, which is why recorded-call announcements are ubiquitous despite not being federally required.

Four limitations should be stated plainly. An indicator is trivially defeatable, though this is true of every notice requirement and has not been thought to vitiate any of them. Notice is not comprehension; a counterparty who sees a light may not know what it implies, and habituation is rapid. The light is useless in remote interactions, which is a substantial coverage gap given how much consequential contracting occurs by telephone and by text. And it does nothing whatever for source monitoring. Latency floors and indicators are therefore complements rather than alternatives, addressing different levels of the framework set out in Part II.

C. Retrospective Detection, and Why This Article Declines It

A logging and retention requirement would serve courts and regulators in a way an indicator cannot. A record of when coupling occurred would support burden-shifting, permit after-the-fact review, and give strict liability something to attach to.

This Article does not recommend it. A continuous record of a coupled interaction is a record of a person's cognition and of every conversation she has, including conversations with people who never consented to the arrangement and cannot see it. Retention creates subpoena exposure, breach exposure, and a standing surveillance infrastructure of unusual intimacy. The doctrinal gain is real but modest, since even a complete record cannot supply the counterfactual that authorship attribution requires, if there is one to supply. The instrument purchases evidence of presence at the price of a permanent record of thought, when an indicator supplies contemporaneous notice of presence without creating the durable cognitive record that logging requires.

D. Allocating What Cannot Be Proven

Some allocation of the unavailable counterfactual is unavoidable, and it is better made by rule than by default. Three approaches are available.

Where coupling is detected, a rebuttable presumption could place the cost of the unresolvable question on the party in the best position to have avoided it. The move is not novel. Corporate attribution already allocates attribution without resolving cognitive authorship: agency does not ask which officer conceived a term before binding the entity to it. It designates whose manifestations count and attributes them accordingly. What the corporate case supplies and the coupled case does not is a party to designate: a presumption keyed to detected coupling fills that gap by rule instead. This does not determine authorship; it decides who bears the consequence of authorship being undeterminable.

Duties could run to the coupled user from the deployer directly, on the model of investment adviser regulation (15 U.S.C. § 80b-6), which already contemplates an interested party supplying judgment and constrains its interest rather than measuring its influence. The couple may be uncountable, but the deployer is not.

And in bounded domains the count can be made one by construction. Coupling-free requirements in the voting booth, in testimony, and at certain moments of consent do not infer how many authors produced a judgment; they guarantee the answer. This instrument assumes the judgment is furnished inside the zone. Where it was furnished over the preceding months, a coupling-free booth guarantees a count that no longer corresponds to anything. Enforcement depends on detection, which is why this instrument presupposes the second.

E. What This Article Does Not Reach

Criminal liability presents the same predicate failure and is not treated here. The reason is not that the question is unimportant but that it is overdetermined: duress, diminished capacity, involuntary act, accomplice liability, and conspiracy each engage a coupled defendant differently, and conspiracy in particular requires agreement between two or more persons, which states the individuation problem in statutory text. A treatment adequate to that doctrine requires more space than this Article can give it.

F. Limits

None of these instruments is complete, and each has limitations. An indicator is defeatable. A latency floor degrades the product, and reaches only the moments a rule designates: it does nothing about judgment furnished across the days or years in which a view forms. Deployer duties invite capture by the parties they regulate. Coupling-free zones are unenforceable without detection. What they have in common is that none requires knowing how many persons produced a judgment, which is the only property that matters, because that is the thing that cannot be known.

References

- [1] Artificial Analysis. 2026. Speech to Speech. <https://artificialanalysis.ai/speech-to-speech>. Accessed August 2026.
- [2] Hui Bai, Jan G. Voelkel, Shane Muldowney, Johannes C. Eichstaedt, and Robb Willer. 2025. LLM-generated messages can persuade humans on policy issues. *Nature Communications* 16 (2025), Article 6037. <https://doi.org/10.1038/s41467-025-61345-5>
- [3] Daniel C. Borges. 2026. Artificial personhood: the implications of recognizing the legal personhood of artificial intelligence. *Vanderbilt Journal of Entertainment and Technology Law* 28, 3 (2026), 411.
- [4] David J. Chalmers. 1995. Absent qualia, fading qualia, dancing qualia. In Thomas Metzinger (ed.), *Conscious Experience*. Ferdinand Schoningh, Paderborn, 309-328.
- [5] Andy Clark and David Chalmers. 1998. The extended mind. *Analysis* 58, 1 (1998), 7-19.
- [6] Thomas H. Costello, Gordon Pennycook, and David G. Rand. 2024. Durably reducing conspiracy beliefs through dialogues with AI. *Science* 385, 6714 (2024), Article eadq1814.
- [7] Hamming AI. 2026. Voice AI latency: what's fast, what's slow, how to fix it. <https://hamming.ai/resources/voice-ai-latency-whats-fast-whats-slow-how-to-fix-it>. Accessed August 2026.
- [8] Lukas Holbling, Sebastian Maier, and Stefan Feuerriegel. 2025. A meta-analysis of the persuasive power of large language models. *Scientific Reports* 15 (2025), Article 43818. <https://doi.org/10.1038/s41598-025-30783-y>
- [9] Maurice Jakesch, Advait Bhat, Daniel Buschek, Lior Zalmanson, and Mor Naaman. 2023. Co-writing with opinionated language models affects users' views. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, Article 111, 15 pages. <https://doi.org/10.1145/3544548.3581196>
- [10] Marcia K. Johnson, Shahin Hashtroudi, and D. Stephen Lindsay. 1993. Source monitoring. *Psychological Bulletin* 114, 1 (1993), 3-28.
- [11] Marcia K. Johnson and Carol L. Raye. 1981. Reality monitoring. *Psychological Review* 88, 1 (1981), 67-85.
- [12] Krishna K. Ladha. 1992. The Condorcet jury theorem, free speech, and correlated votes. *American Journal of Political Science* 36, 4 (1992), 617-634.
- [13] Krishna K. Ladha. 1993. Condorcet's jury theorem in light of de Finetti's theorem: majority-rule voting with correlated votes. *Social Choice and Welfare* 10, 1 (1993), 69-86.
- [14] Sandra C. Matz, Jacob D. Teeny, Sumer S. Vaid, Heinrich Peters, Gabriella M. Harari, and Moran Cerf. 2024. The potential of generative AI for personalized persuasion at scale. *Scientific Reports* 14 (2024), Article 4692.
- [15] Karen J. Mitchell and Marcia K. Johnson. 2009. Source monitoring 15 years later. *Psychological Bulletin* 135, 4 (2009), 638-677.
- [16] Sander Noels, Alexander Rogiers, Maarten Buyl, and Tijl De Bie. 2026. Persuasion with large language models: a survey of empirical evidence, study methodologies, and ethical implications. [arXiv:2411.06837v2](https://arxiv.org/abs/2411.06837v2).

- [17] OpenAI. 2024. Hello GPT-4o. <https://openai.com/index/hello-gpt-4o/>.
- [18] OpenBenchmarks. 2026. Voice agent end-to-end latency. <https://openbenchmarks.com/voice-agent-latency/voice-agent-end-to-end-latency>. Accessed August 2026.
- [19] Luca Ronconi, Nikolaas N. Oosterhof, Claudia Bonmassar, and David Melcher. 2017. Multiple oscillatory rhythms determine the temporal organization of perception. *Proceedings of the National Academy of Sciences* 114, 51 (2017), 13435-13440.
- [20] Susan Schneider. 2019. *Artificial You: AI and the Future of Your Mind*. Princeton University Press, Princeton, NJ.
- [21] Daniel Thilo Schroeder, Meeyoung Cha, Andrea Baronchelli, Nick Bostrom, Nicholas A. Christakis, David Garcia, Amit Goldenberg, Yara Kyrychenko, Kevin Leyton-Brown, Nina Lutz, Gary Marcus, Filippo Menczer, Gordon Pennycook, David G. Rand, Maria Ressa, Frank Schweitzer, Dawn Song, Christopher Summerfield, Audrey Tang, Jay J. Van Bavel, Sander van der Linden, and Jonas R. Kunst. 2026. How malicious AI swarms can threaten democracy. *Science* 391, 6783 (2026), 354-357. <https://doi.org/10.1126/science.adz1697>
- [22] Tanya Stivers, N. J. Enfield, Penelope Brown, et al. 2009. Universals and cultural variation in turn-taking in conversation. *Proceedings of the National Academy of Sciences* 106, 26 (2009), 10587-10592. <https://doi.org/10.1073/pnas.0903616106>
- [23] Abel Wajnerman-Paz, Mario Villalobos, and Juan Fuentes. 2026. Autopoietic bodily integrity: a biological approach to hybrid minds. *Minds and Machines* 36 (2026), Article 37. <https://doi.org/10.1007/s11023-026-09792-x>
- [24] Daniel M. Wegner and Thalia Wheatley. 1999. Apparent mental causation: sources of the experience of will. *American Psychologist* 54, 7 (1999), 480-492.
- [25] Miguel A. Hernan. 2016. Does water kill? A call for less casual causal inferences. *Annals of Epidemiology* 26, 10 (2016), 674-680. <https://doi.org/10.1016/j.annepidem.2016.08.016>
- [26] Paul W. Holland. 1986. Statistics and causal inference. *Journal of the American Statistical Association* 81, 396 (1986), 945-960. <https://doi.org/10.1080/01621459.1986.10478354>

Legal Authorities

Cases

- Lujan v. Defenders of Wildlife, 504 U.S. 555 (1992).
- Odorizzi v. Bloomfield School District, 246 Cal. App. 2d 123 (1966).
- Re A (Children) [2001] Fam 147 (CA).
- Reynolds v. Sims, 377 U.S. 533 (1964).
- Sterling Drug, Inc. v. Cornish, 370 F.2d 82 (8th Cir. 1966).
- Thornburg v. Gingles, 478 U.S. 30 (1986).
- Wesberry v. Sanders, 376 U.S. 1 (1964).

Statutes and Regulations

Cal. Penal Code § 632.

FTC Cooling-Off Rule, 16 C.F.R. pt. 429.

Investment Advisers Act of 1940, 15 U.S.C. § 80b-6.

Truth in Lending Act, 15 U.S.C. § 1635.

Voting Rights Act § 208, 52 U.S.C. § 10508.

Voting Rights Act § 2, 52 U.S.C. § 10301

18 U.S.C. § 2511.

Other Authorities

1 William Blackstone, Commentaries on the Laws of England *430 (1765).

Restatement (Second) of Contracts §§ 12-16 (1981) (capacity).

Restatement (Second) of Contracts § 177 (1981) (undue influence; § 177(3), third-party undue influence).

Restatement (Second) of Contracts § 208 (1981) (unconscionable contract or term).

Restatement (Third) of Agency § 1.01 (2006) (definition of agency).

Restatement (Third) of Agency § 2.03 (2006) (apparent authority).